

1.1 Machine Learning Basics

Supervised, unsupervised, batch versus online, and the usual failure modes.

1. Four learning setups

This course is advanced machine learning. The algorithms change. The **setup** usually does not.

Supervised. The training rows come with answers (labels). Classification puts a row in a bucket (spam / ham). Regression predicts a number (price of a car). Methods you already know still apply: (k)-NN, linear and logistic regression, SVMs, trees, forests, neural nets. Weeks 5 and 12–15 pick these up again with more math.

Unsupervised. No labels. Clustering ((k)-means, hierarchical), anomaly detection (one-class SVM, later GMMs), and latent-variable models (PCA, kernel PCA, ICA, tensors) are the second half of the semester.

Semi-supervised. A little labeled data, a lot of unlabeled. A photo app that clusters faces and then asks you to name each cluster is the usual picture.

Reinforcement. An agent takes actions, sees a reward, and updates a policy. We will not live here. Know the words.

The Géron book is the informal source for this lecture. *Mathematics for Machine Learning* is the math source for the rest of Module 1.

2. Batch versus online

Batch (offline). Fit on the whole training set, then freeze the model. Simple. Expensive if you retrain from scratch every night on a huge matrix. Fine when the world is slow.

Online. Update as points arrive. The **learning rate** is how fast you forget the past. High: you track a changing stream and you also chase noise. Low: you are stable and you are late.

A live system that trains online can be poisoned by one bad sensor. Monitor the inputs. Be ready to switch learning off.

3. Why models fail

Problem	What it looks like	What you do
Too little data	High variance, lucky splits	More data, or a simpler model
Non-representative data	Sampling bias; the test world is not the train world	Fix how you sample
Poor quality	Errors, outliers, sensor junk	Clean before you model
Irrelevant features	The signal is not in the columns	Select, extract, or collect better ones
Underfitting	Train and test both bad	Richer model, better features, less regularization
Overfitting	Train great, test poor	Simpler model, more data, less noise

Data can beat a clever algorithm on a large problem. Medium data is still the usual case in this course. Do not abandon the math.